

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1915

Almena 100 kV
November 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1915. The Interconnection Customer proposes a 123.3 MW solar photovoltaic facility interconnecting at the Almena 100 kV substation in the Michigan area. The request was received on October 27, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 652 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$981,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Almena / Portola 100kV
Contingency: Oceanside / Palisades 500kV line
Loading: 114.5% (Rating: 141 MVA)
DFAX: 69.3%
Pre-project loading: 53.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Almena / Portola 100kV L	Rebuild line with higher-capacity conductor	\$1,534,000	64% (\$981,000)
	Total allocated to project		\$981,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.